Fires and Explosions

Chemicals present a substantial hazard in the form of fires and explosions. The combustion of one gallon of toluene can destroy an ordinary chemistry laboratory in minutes; persons present may be killed. The potential consequences of fires and explosions in pilot plants and plant environments are even greater.

The three most common chemical plant accidents are fires, explosions, and toxic releases, in that order (see Chapter 1). Organic solvents are the most common source of fires and explosions in the chemical industry.

Chemical and hydrocarbon plant losses resulting from fires and explosions are substantial, with yearly property losses in the United States estimated at almost $300 million (1997 dollars).[1] Additional losses in life and business interruptions are also substantial. To prevent accidents resulting from fires and explosions, engineers must be familiar with

- The fire and explosion properties of materials,
- The nature of the fire and explosion process, and
- Procedures to reduce fire and explosion hazards.

The Fire Triangle

The essential elements for combustion are fuel, an oxidizer, and an ignition source. These elements are illustrated by the fire triangle, shown in Figure 6-1.

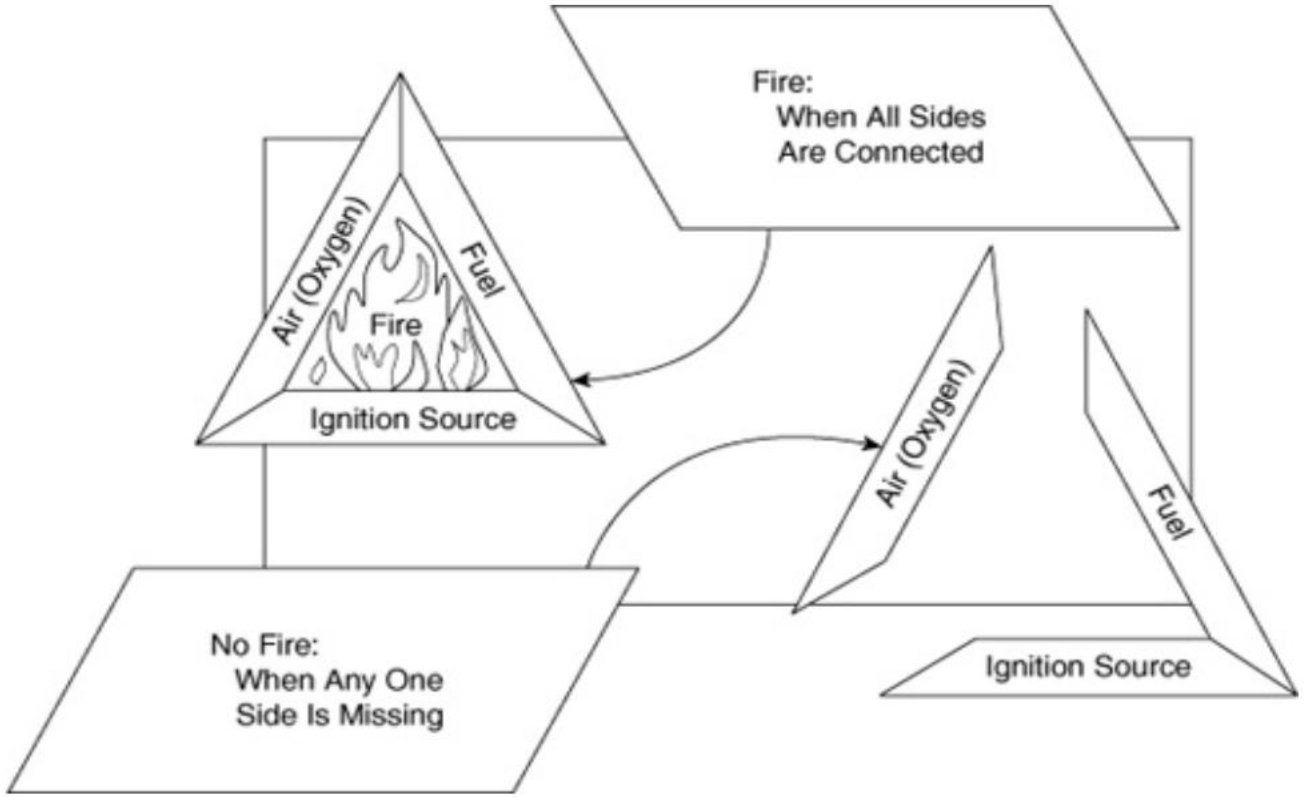

Figure 6-1. The fire triangle.

Fire, or burning, is the rapid exothermic oxidation of an ignited fuel. The fuel can be in solid, liquid, or vapor form, but vapor and liquid fuels are generally easier to ignite. The combustion always occurs in the vapor phase; liquids are volatized and solids are

decomposed into vapor before combustion.

When fuel, oxidizer, and an ignition source are present at the necessary levels, burning will occur. This means a fire will *not* occur if (1) fuel is not present or is not present in sufficient quantities, (2) an oxidizer is not present or is not present in sufficient quantities, and (3) the ignition source is not energetic enough to initiate the fire.

Two common examples of the three components of the fire triangle are wood, air, and a match; and gasoline, air, and a spark. However, other, less obvious combinations of chemicals can lead to fires and explosions. Various fuels, oxidizers, and ignition sources common in the chemical industry are

Fuels

Liquids: gasoline, acetone, ether, pentane

Solids: plastics, wood dust, fibers, metal particles

Gases: acetylene, propane, carbon monoxide, hydrogen

Oxidizers

Gases: oxygen, fluorine, chlorine

Liquids: hydrogen peroxide, nitric acid, perchloric acid

Solids: metal peroxides, ammonium nitrite

Ignition sources

Sparks, flames, static electricity, heat

Distinction between Fires and Explosions

The major distinction between fires and explosions is the rate of energy release. Fires release energy slowly, whereas explosions release energy rapidly, typically on the order of microseconds. Fires can also result from explosions, and explosions can result from fires.

A good example of how the energy release rate affects the consequences of an accident is a standard automobile tire. The compressed air within the tire contains energy. If the energy is released slowly through the nozzle, the tire is harmlessly deflated. If the tire ruptures suddenly and all the energy within the compressed tire releases rapidly, the result is a dangerous explosion.

Some of the commonly used definitions related to fires and explosions are given

Combustion or fire: Combustion or fire is a chemical reaction in which a substance combines with an oxidant and releases energy. Part of the energy released is used to sustain the reaction.

Ignition: Ignition of a flammable mixture may be caused by a flammable mixture coming in contact with a source of ignition with sufficient energy or the gas reaching a temperature high enough to cause the gas to autoignite.

Autoignition temperature (AIT): A fixed temperature above which adequate energy is available in the environment to provide an ignition source.

Flash point (FP): The flash point of a liquid is the lowest temperature at which it gives off enough vapor to form an ignitable mixture with air. At the flash point the vapor will burn but only briefly; inadequate vapor is produced to maintain combustion. The flash point generally increases with increasing pressure.

There are several different experimental methods used to determine flash points. Each method produces a somewhat different value. The two most commonly used methods are open cup and closed cup, depending on the physical configuration of the experimental equipment. The open-cup flash point is a few degrees higher than the closed-cup flash point.

Fire point: The fire point is the lowest temperature at which a vapor above a liquid will continue to burn once ignited; the fire point temperature is higher than the flash point.

Flammability limits: Vapor-air mixtures will ignite and burn only over a well-specified range of compositions. The mixture will not burn when the composition is lower than the lower flammable limit (LFL); the mixture is too lean for combustion. The mixture is also not combustible when the composition is too rich, that is, when it is above the upper flammable limit (UFL). A mixture is flammable only when the composition is between the LFL and the UFL. Commonly used units are volume percent fuel (percentage of fuel plus air).

Lower explosion limit (LEL) and upper explosion limit (UEL) are used interchangeably with LFL and UFL.

Explosion: An explosion is a rapid expansion of gases resulting in a rapidly moving pressure or shock wave. The expansion can be mechanical (by means of a sudden rupture of a pressurized vessel), or it can be the result of a rapid chemical reaction. Explosion damage is caused by the pressure or shock wave.

Mechanical explosion: An explosion resulting from the sudden failure of a vessel containing high-pressure nonreactive gas.

Deflagration: An explosion in which the reaction front moves at a speed less than the speed of sound in the unreacted medium.

Detonation: An explosion in which the reaction front moves at a speed greater than the speed of sound in the unreacted medium.

Confined explosion: An explosion occurring within a vessel or a building. These are most common and usually result in injury to the building inhabitants and extensive damage.

Unconfined explosion: Unconfined explosions occur in the open. This type of explosion is usually the result of a flammable gas release. The gas is dispersed and mixed with air until it comes in contact with an ignition source. Unconfined explosions are rarer than confined explosions because the explosive material is frequently diluted below the LFL by wind dispersion. These explosions are destructive because large quantities of gas and large areas are frequently involved.

Boiling-liquid expanding-vapor explosion (BLEVE): A BLEVE occurs if a vessel that contains a liquid at a temperature above its atmospheric pressure boiling point ruptures. The subsequent BLEVE is the explosive vaporization of a large fraction of the vessel contents, possibly followed by combustion or explosion of the vaporized cloud if it is combustible. This type of explosion occurs when an external fire heats the contents of a tank of volatile material. As the tank contents heat, the vapor pressure of the liquid within the tank increases and the tank's structural integrity is reduced because of the heating. If the tank ruptures, the hot liquid volatilizes explosively.

Dust explosion: This explosion results from the rapid combustion of fine solid particles. Many solid materials (including common metals such as iron and aluminum) become flammable when reduced to a fine powder.

Shock wave: An abrupt pressure wave moving through a gas. A shock wave in open air is followed by a strong wind; the combination of shock wave and wind is called a blast wave. The pressure increase in the shock wave is so rapid that the process is mostly adiabatic.

Overpressure: The pressure on an object as a result of an impacting shock wave.

Explosion behavior depends on a large number of parameters. A summary of the more important parameters is shown in Table 6-6.

Table 6-6. Parameters Significantly Affecting the Behavior of Explosions

Ambient temperature
Ambient pressure
Composition of explosive material
Physical properties of explosive material
Nature of ignition source: type, energy, and duration
Geometry of surroundings: confined or unconfined
Amount of combustible material
Turbulence of combustible material
Time before ignition
Rate at which combustible material is released

Explosion behavior is difficult to characterize. Many approaches to the problem have been undertaken, including theoretical, semiempirical, and empirical studies. Despite these efforts, explosion behavior is still not completely understood. Practicing engineers, therefore, should use extrapolated results cautiously and provide a suitable margin of safety in all designs.

Vapor Cloud Explosions

The most dangerous and destructive explosions in the chemical process industries are vapor cloud explosions (VCEs). These explosions occur in a sequence of steps:

1. Sudden release of a large quantity of flammable vapor (typically this occurs when a vessel, containing a superheated and pressurized liquid, ruptures),
2. Dispersion of the vapor throughout the plant site while mixing with air, and
3. Ignition of the resulting vapor cloud.

The accident at Flixborough, England, is a classic example of a VCE. A sudden failure of a 20-inch cyclohexane line between reactors led to vaporization of an estimated 30 tons of cyclohexane. The vapor cloud dispersed throughout the plant site and was ignited by an unknown source 45 seconds after the release. The entire plant site was leveled and 28 people were killed.

Boiling-Liquid Expanding-Vapor Explosions

A boiling-liquid expanding-vapor explosion (BLEVE, pronounced ble′-vee) is a special type of accident that can release large quantities of materials. If the materials are flammable, a VCE might result; if they are toxic, a large area might be subjected to toxic materials. For either situation the energy released by the BLEVE process itself can result in considerable damage.

A BLEVE occurs when a tank containing a liquid held above its atmospheric pressure boiling point ruptures, resulting in the explosive vaporization of a large fraction of the tank contents.